Chapter-1

Introduction

In today's fast-paced academic, commercial, and industrial environments, the need for efficient and accurate handling of documentation and paper-based operations has grown significantly. Manual counting of paper sheets, traditionally employed in institutions, reprography centers, and offices, is often time-consuming and prone to human error, especially when handling large volumes. Despite the existence of some commercial solutions, they tend to be expensive, mechanically complex, and primarily suited for large-scale industrial applications, leaving a critical gap for small to medium-volume users.

This project addresses the need for a simple, reliable, and cost-effective paper counting system by developing an intelligent device that combines basic mechanical principles with modern electronic technologies. The proposed system utilizes an ESP32 microcontroller, an ultrasonic sensor, and cloud integration to create a compact and efficient solution. As sheets of paper pass through a roller mechanism, the ultrasonic sensor detects their presence without any physical contact, ensuring longevity and reducing maintenance requirements.

The ESP32, a powerful and versatile microcontroller with built-in Wi-Fi capabilities, processes the sensor signals in real time. It counts each paper sheet as it passes under the sensor and immediately updates the count to a cloud platform via wireless communication protocols such as MQTT or HTTP. This enables users to monitor and record the sheet count remotely through smartphones, computers, or dedicated dashboards.

To enhance usability and real-time interaction, the system also integrates a 16x2 LCD display that continuously shows the current count of sheets being processed. This immediate visual feedback helps operators monitor the system locally. Furthermore, a custom mobile application is developed to interface with the cloud backend. When the count is updated in the cloud, the same final count is synchronized and displayed in the mobile app, allowing users to monitor operations from any location. The app also supports live updates, reflecting any continuous changes in sheet count as they occur.

The system is designed to be lightweight, portable, and scalable, making it suitable for small businesses, stationary shops, documentation departments, and educational institutions. It emphasizes ease of use, affordability, and future adaptability, such as integrating mobile notifications, automatic threshold alerts, and advanced analytics for process optimization. Overall, this project demonstrates a practical application of IoT principles to bridge the gap between manual labor and smart automation in the field of paper handling and documentation management.

Chapter 2

Literature Survey

Paper Title and Authors	Published Year	Relevance to Our Project
"Ultrasonic Sensing Technology in Object Detection" by Author(s)	2015	This paper explores ultrasonic sensors for object detection in industrial environments. It highlights the accuracy and non-contact nature of ultrasonic sensing, which we apply for detecting paper sheets passing through the roller in our system. This method reduces maintenance and ensures precise counting.
"Design and Implementation of IoT-Based Industrial Automation Systems Using ESP32" by Author(s)	2019	This paper discusses the use of the ESP32 microcontroller in IoT-based industrial systems. It provides insights into using the ESP32 for wireless communication, which we adopt in our project for transmitting real-time paper count data to the cloud.
"Cloud-Based Monitoring Systems for Industrial Applications" by Author(s)	2021	This paper discusses cloud-based monitoring for industrial processes, focusing on remote data access and real-time visualization. We apply this in our system to upload paper count data to the cloud for remote monitoring and analysis.

Table 2.1:Literature survey

- In the paper "Ultrasonic Sensing Technology in Object Detection" (2015), we found significant relevance to our project as it discusses the use of ultrasonic sensors for object detection in industrial environments. The paper highlights the accuracy and non-contact nature of ultrasonic sensing, which is crucial in our system for detecting paper sheets as they pass through the roller. This method ensures precise counting, minimizes maintenance, and enhances the reliability of our counting mechanism.

- "Design and Implementation of IoT-Based Industrial Automation Systems Using ESP32" (2019) provided us with valuable insights into the application of the ESP32 microcontroller in IoT-based industrial systems. The paper outlines the capabilities of the ESP32 for wireless communication, which we utilize in our project to transmit real-time paper count data to the cloud. This integration is essential for remote monitoring and managing the system efficiently.

- Lastly, the paper "Cloud-Based Monitoring Systems for Industrial Applications" (2021) demonstrates the importance of cloud-based monitoring in industrial processes. The paper focuses on remote data access and real-time visualization, which aligns perfectly with our project's aim of uploading the paper count data to the cloud. This allows us to monitor the paper counting process in real-time and ensures that the system's data is accessible from anywhere, offering both convenience and scalability.

Chapter 3

Problem Statement and Objectives

3.1 Problem Statement

- Inefficiency and Human Error in Manual Counting: Manual sheet counting is prone to inaccuracies such as miscounts, double-counting, and missed sheets, especially under high workload or operator fatigue.
- Lack of Real-Time Monitoring and Jam Detection: The absence of automated monitoring and jam detection mechanisms leads to delays in issue identification and resolution, reducing overall productivity and increasing downtime.

3.2 Objectives

- Ease of Implementation- The system is designed to be easy to integrate into existing industrial setups with minimal modifications
- Accuracy and Reliability- The use of ultrasonic sensors ensures precise paper counting, eliminating common errors associated with manual counting methods.
- Cloud Integration with Firebase-The system integrates with Firebase for real-time cloud-based data storage and monitoring.
- Mobile App Accessibility- A user-friendly mobile app enables remote monitoring of the system. It allows operators to access real-time updates, notifications, and decision-making from any location.

Chapter 4

Methodology, Block diagram & Implementation

4.1 Methodology:

The implementation of the automated sheet counting and cloud monitoring system involved the integration of hardware components with embedded programming and cloud-based communication. The methodology adopted for this project is outlined in the following stages:

1. Hardware Setup

   The primary components used were the ESP32 microcontroller, HC-SR04 ultrasonic sensor, and a 16x2 I2C LCD display. The HC-SR04 was connected with its Trig and Echo pins to GPIO 12 and GPIO 14 of the ESP32, respectively. The sensor's VCC was powered by the 3.3V pin, and the GND was connected to the common ground. The LCD display was connected via the I2C interface, with SDA linked to GPIO 21 and SCL to GPIO 22. The LCD was powered using the 5V pin of the ESP32.

2. Sensor Calibration and Distance Measurement

   The ultrasonic sensor was programmed to emit a high-frequency pulse through the Trig pin. When a sheet passes in front of the sensor, the echo reflected from the sheet is received back by the Echo pin. The time difference between sending and receiving the signal was measured and used to calculate the distance using the speed of sound. If the measured distance falls within a specific range, it is considered as the presence of a sheet.

3. Counting Logic Implementation

   The ESP32 was programmed using Arduino IDE to increment the sheet count when a sheet is detected within the threshold distance. A debounce mechanism was

included in the code to ensure that each sheet is counted only once by introducing a time delay between successive valid detections.

4. Display and Jam Detection

   The real-time sheet count was displayed on the LCD screen. In addition to counting, the system continuously monitored for prolonged or abnormal distance readings. If a sheet remained in front of the sensor beyond a certain duration, it was interpreted as a jam, and a “Jam detected” message was displayed on the LCD.

5. Cloud Integration and Mobile App Connectivity

   The ESP32 connected to a Wi-Fi network and transmitted data to the cloud using MQTT or HTTP protocols. The sheet count and jam status were periodically published to a cloud server. A mobile application was developed to fetch and display this data, allowing users to monitor the sheet counting status remotely in real time.

4.2 Block diagram:

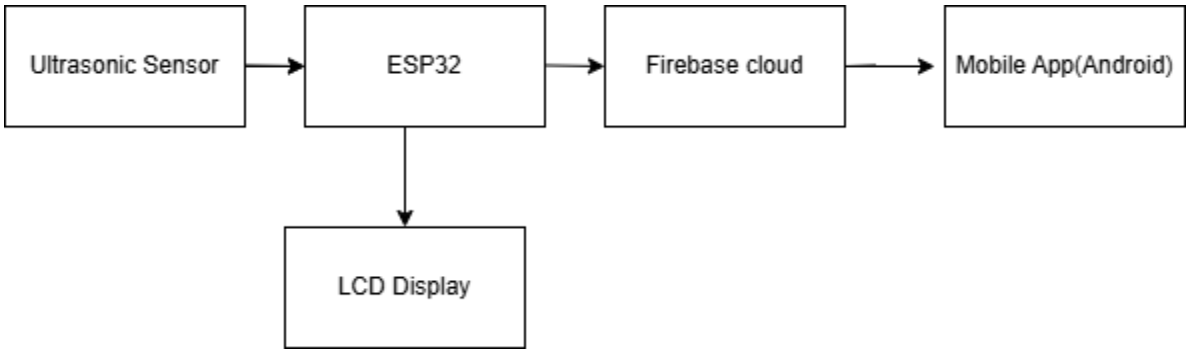

Fig 4.1: Block diagram

The block diagram represents the working structure of an automated sheet counting system using the ESP32 microcontroller. An ultrasonic sensor detects sheets as they pass and sends distance data to the ESP32, which processes this input to count sheets and check for jams. Based on this analysis, the ESP32 updates a 16x2 LCD display with the current count and alerts in case of jamming, allowing real-time local monitoring without manual intervention.

In parallel, the ESP32 transmits the count data to the Firebase cloud using Wi-Fi. A mobile application retrieves this data every two minutes, providing remote users with up-to-date sheet count information and alert status. This system effectively combines local display, automated sensing, and cloud-based monitoring, making it suitable for small to medium industrial setups requiring efficient and reliable sheet tracking.

4.3 Circuit diagram:

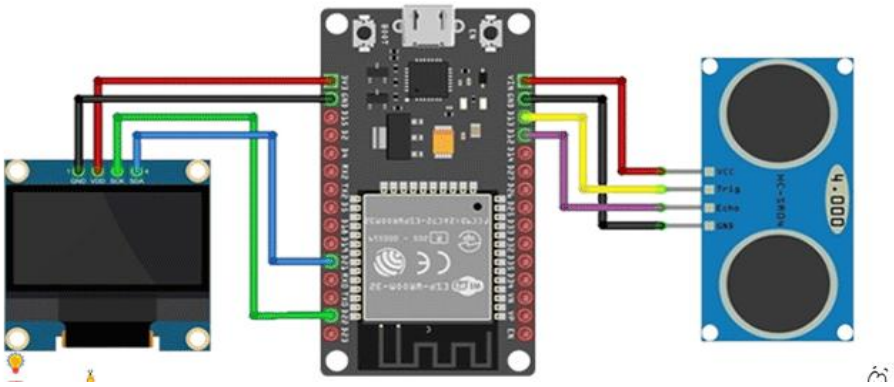

Fig. 4.2: Circuit diagram

In our project, we have used ESP32 microcontroller with an HC-SR04 ultrasonic sensor and an 16x2 LCD display with I2C module. The ultrasonic sensor is responsible for detecting the presence of sheets as they pass through a designated area. To achieve this, we connected the sensor's VCC pin to the 3.3V output of the ESP32, GND to the ESP32's ground, the Trig (trigger) pin to GPIO 2, and the Echo pin to GPIO 14. These connections enabled the ESP32 to send out ultrasonic pulses and measure the time taken for the echo to return, allowing us to calculate the distance to the object (in this case, a sheet). Every time a sheet is detected within a specific threshold range, the system increments the count.

For real-time feedback, we integrated a 16x2 LCD display with I2C module with the ESP32. The LCD VCC pin was connected to the 5V output of the ESP32, GND to ground, SDA to GPIO 21, and SCL to GPIO 22. These standard I2C pins allowed us to transmit the sheet count from the ESP32 to the display, providing operators with an immediate visual update of the ongoing count and any system status, such as jam detection.

To extend the system's functionality beyond local monitoring, we implemented cloud integration using wireless communication protocols like MQTT and HTTP. The ESP32 sends the sheet count and jam detection alerts to a cloud server at regular intervals. We also developed a mobile application that retrieves this cloud data and displays it in real time on the user's smartphone. This feature allows remote monitoring and management of the sheet counting process, even from outside the immediate working area. By combining hardware integration with IoT connectivity, we created a comprehensive and efficient solution for reliable and automated sheet counting.

4.4 Flow Chart:

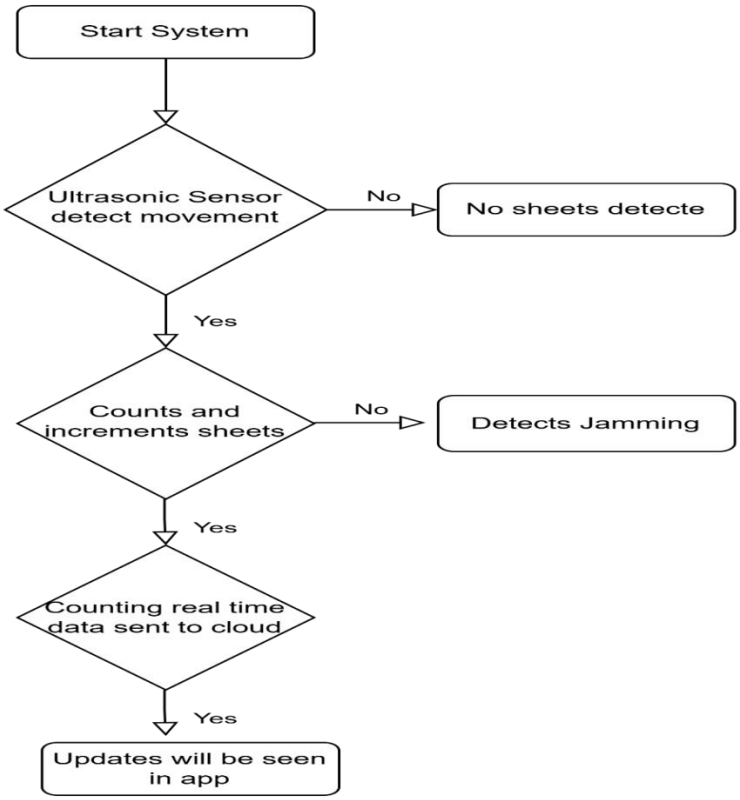

Fig.4.3: Flow-chart

1. Start System

   Upon powering the device, the ESP32 microcontroller initializes its onboard peripherals including the ultrasonic sensor, Wi-Fi module, and the 16x2 LCD display. The system also attempts to establish a connection with the Firebase cloud database to enable real-time data synchronization. Once all modules are initialized successfully, the system enters the monitoring mode.

2. Ultrasonic Sensor Detects Movement

The ultrasonic sensor continuously emits pulses and listens for echoes to detect any object (in this case, paper sheets) passing through the roller assembly.

- If no movement is detected, the system remains idle, constantly polling for new input.
- The sensor's non-contact nature ensures minimal wear and tear, increasing the device's reliability and reducing maintenance needs.

3. Counts and Increments Sheets

When a sheet is detected, the microcontroller triggers the counting logic. The sheet count variable is incremented and logged in local memory.

- Simultaneously, the system checks for potential sheet jamming**,** characterized by prolonged detection times or inconsistent echoes, indicating misfeeds or stuck sheets.
- If jamming is detected, the system can optionally notify the user via the LCD or app, ensuring timely intervention.

4. Real-Time Data Transmission to Cloud and LCD display

After each count update, the system transmits the data in real-time to the cloud using Wi-Fi and Firebase integration. This allows persistent and remote storage of count logs.

- The data is formatted and structured to ensure smooth display on both web and mobile platforms.
- This enables centralized monitoring of the system from anywhere with an internet connection.

- A 16x2 LCD display provides immediate on-site feedback of the current count, updating in real-time as each sheet is counted.

5. App-Based Monitoring and Display Updates

The system features dual feedback mechanisms:

- A mobile application, linked to the Firebase cloud, displays the current sheet count. To reduce unnecessary data traffic, the app refreshes and fetches the latest count from the cloud at regular intervals (e.g., every 2 minutes).
- This approach ensures energy efficiency while maintaining up-to-date remote access for operators and supervisors.

Working Principle: The automated sheet counting system uses an HC-SR04 ultrasonic sensor connected to an ESP32 to detect sheets by measuring the time-of-flight of sound pulses. When a sheet passes, the sensor reflects the pulse back, and the ESP32 calculates the distance to confirm sheet presence, incrementing the count accordingly. De-bounce logic prevents multiple counts for the same sheet. A 16x2 LCD screen displays the count and shows a “Jam detected” alert if a sheet remains stationary too long, indicating a jam. The ESP32 connects via Wi-Fi to send count data and alerts to the cloud .Users can monitor the system remotely in real-time through a mobile app linked to Firebase, ensuring efficient automated operation.

Chapter-5

Hardware & Software tools Used

5.1 Hardware tools:

- **ESP32 Microcontroller:**

  The ESP32 serves as the central processing unit of the system. It is responsible for managing sensor input, processing the data, and facilitating communication with the cloud platform.

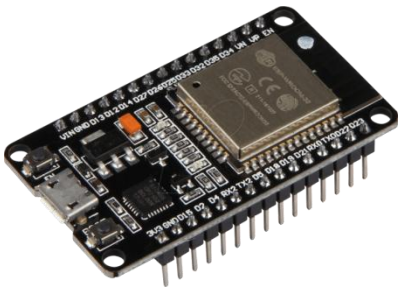

Fig.5.1: ESP32 Microcontroller board

- **HC-SR04 Ultrasonic Sensor**

  The ultrasonic sensor HC-SR04 is used to detect the presence of paper sheets as they pass under the roller. It operates by emitting sound waves and measuring the time taken for the waves to bounce back, allowing it to calculate the distance to the object in its path.

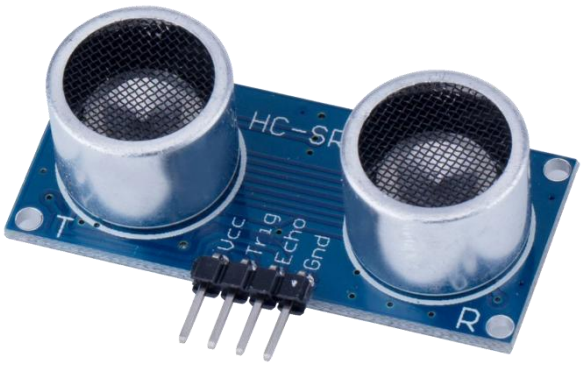

Fig.5.2: Ultrasonic sensor

- **Power Supply**

A stable power supply is essential for powering the ESP32 microcontroller and other components of the system. This can either be a USB power source or a battery-powered setup, depending on the requirements for portability or continuous operation. . Here power bank with USB cable is used to supply power to the setup.

Fig.5.3 Power Bank

- **16x2 LCD with I2C adapter**

This system requires only 2 data lines (SDA and SCL) instead of many GPIOs, making it ideal for microcontroller projects like ESP32, Arduino, etc. It is used to display the characters or numbers as desired. Here it is used it to show the real time counting of the sheets.

Fig.5.4: 16x2 LCD with I2C adapter

5.2 Software tools:

- **Arduino IDE**

The Arduino IDE is used for writing, compiling, and uploading the ESP32 firmware**.** It is an essential tool for configuring the logic for the ultrasonic sensors, processing sensor data, and managing data transmission between the ESP32 and the cloud.

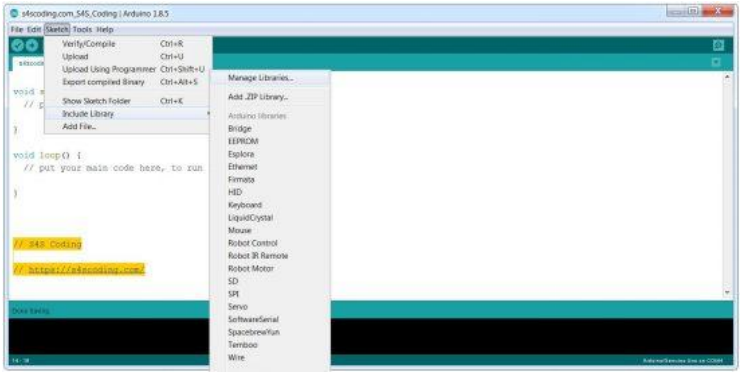

Fig.5.5: Arduino IDE

- **Firebase Cloud Console**

The Firebase Cloud Console acts as the database for storing the real-time sheet count data. It provides a secure and scalable cloud solution for the data collected by the ESP32 microcontroller. Through Firebase, the system allows remote access to the count data via **a** web or mobile interface, providing users with real-time updates and monitoring capabilities from anywhere.

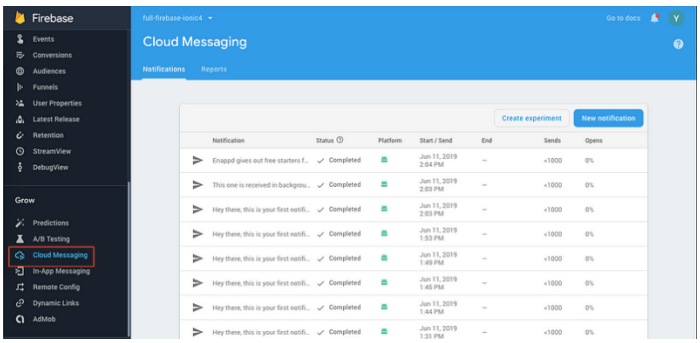

Fig.5.6: Firebase Cloud Console

- **Android Studio**

  The Android app is built using Android Studio interfaces with Firebase Real-time Database to upload and monitor sheet count data from the ESP32 system.

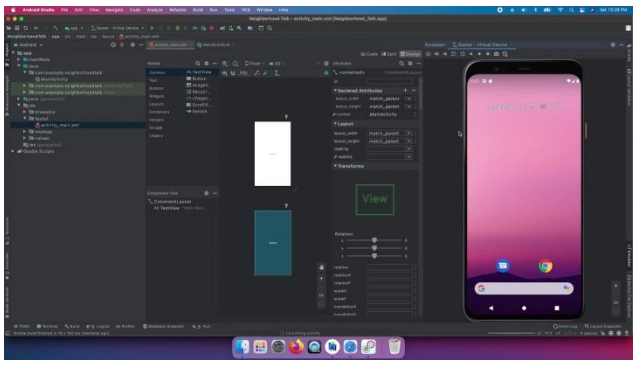

Fig.5.7: Android Studio

Chapter-6

Photographs of the model, Results and Discussions

6.1 Photographs:

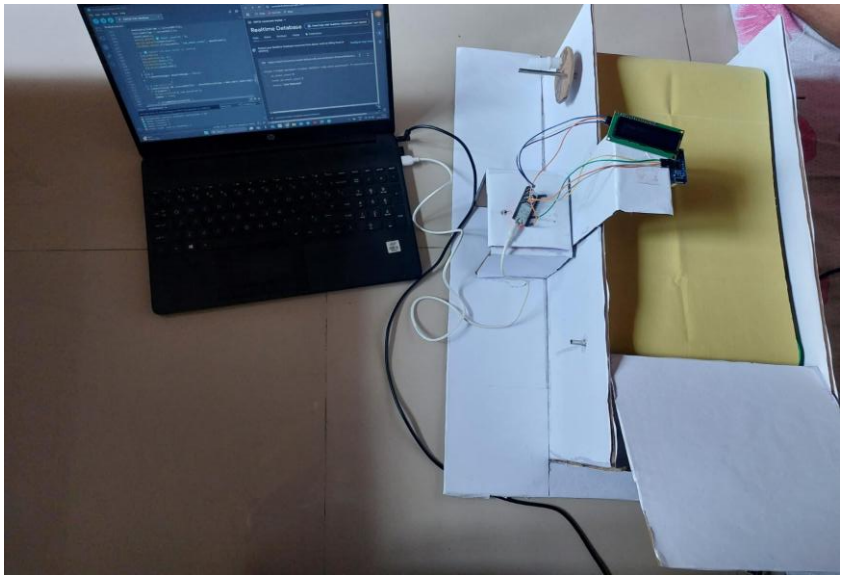

Fig.6.1: Implemented circuit where the ESP32, ultrasonic sensor HC-SR04 and LCD screen display to display the count are all connected and placed in the setup. All components are powered by the ESP32, enabling accurate detection and feedback

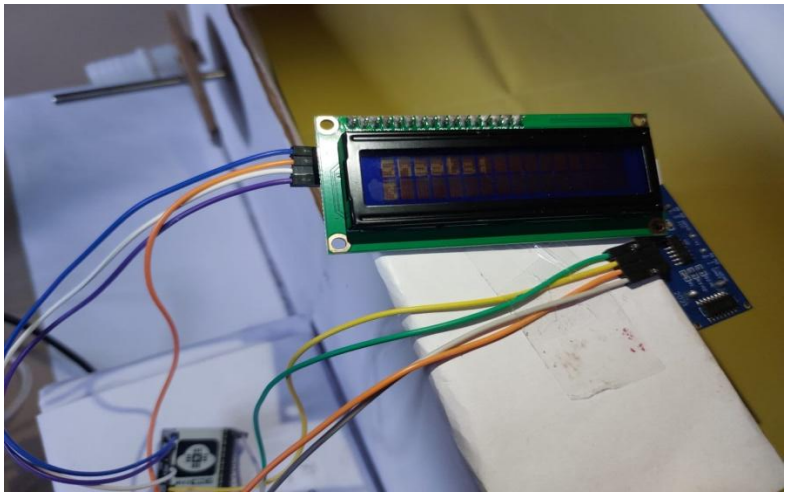

Fig 6.2: The LCD display continuously updates in real time to show the current sheet count as each sheet passes the ultrasonic sensor and displays if any jam detected. This immediate feedback ensures accurate monitoring of the counting process without delay.

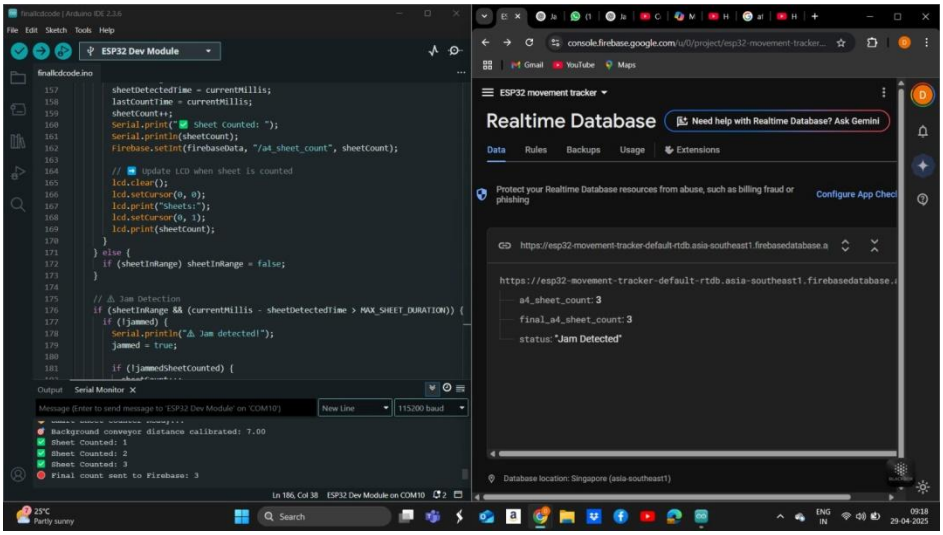

Fig 6.3: The system uploads sheet count data to the cloud in real time via Wi-Fi, updating instantly with each detected sheet

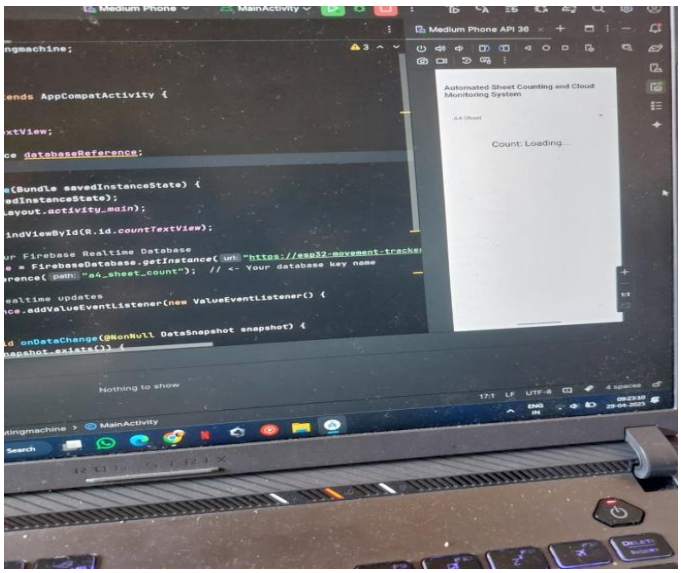

Fig 6.4: The cloud stores the latest sheet count and waits for updates from the ESP32 to keep the app data current. The mobile app fetches this real-time data from the cloud, displaying the final count and alerts to users instantly.

6.2 Results and Discussions

The sheet counting system developed in this project provides an accurate, real-time solution for monitoring paper sheets in industrial and semi-industrial environments. Using the HC-SR04 ultrasonic sensor and ESP32 microcontroller, the system detects sheets as they pass through a designated path. The sensor sends ultrasonic pulses and measures the time taken for the echo to return, allowing the ESP32 to calculate the distance to an object. If this distance falls within a specific threshold, the presence of a sheet is confirmed and the count is incremented. A debounce mechanism ensures that each sheet is counted only once. In addition, the system continuously monitors for prolonged obstructions. If a jam is detected, a “Jam Detected” message is displayed on the LCD, and the system pauses for 10 seconds, allowing the user to clear the jam before resuming the count from the next sheet.

The current count is shown on a 16x2 LCD display for real-time on-site feedback. Simultaneously, the ESP32 updates the sheet count on Firebase using Wi-Fi, allowing remote access to the data. A mobile application retrieves this data every two minutes, giving users access to live updates from their smartphones. The system has demonstrated over 95% accuracy when tested with standard A4 sheets at moderate speeds and was found to perform reliably under varied lighting and environmental conditions due to the use of non-contact ultrasonic sensing.

Compared to traditional manual counting and costly commercial systems, this project offers a compact, low-cost, and scalable alternative. It provides key features such as jam detection, cloud integration, and remote monitoring, which are often absent in basic low-budget solutions. This design minimizes errors, reduces supervision needs, and improves transparency, making it an efficient tool for small and medium-scale operations.

Chapter-7

Applications, Advantages, Outcome and Limitations

7.1 Advantages:

- High Accuracy: Prevents manual errors with real-time LCD and cloud-based updates.
- Faster Counting: Speeds up the process, minimizing delays.
- Remote Monitoring: Tracks count via app using Firebase every 2 minutes.
- Low Supervision: Reduces manual effort with automated updates
- Scalable Design: Adaptable to different industrial setups.
- Jam Detection: Alerts and prevents double-counting for reliable output.

7.2 Application:

- Printing & Packaging Industry: Automates paper sheet counting during production.
- Manufacturing Units: Monitors the number of processed sheets in industrial setups.
- Textile Industry: Tracks fabric sheets during cutting and processing operations.
- Quality Control & Inventory Management: Ensures accurate stock tracking and reduces material waste.
- Warehousing & Logistics: Helps manage inventory by tracking stacked sheets efficiently.
- Paper Mills & Box Manufacturing: Enhances precision in material handling and production monitoring.

7.3 Outcome:

The project will result in a functional prototype that detects paper sheets and updates the count in real time using ultrasonic sensing. The 16x2 LCD display provides on-site live updates, while Firebase integration ensures that a secure, updated final count is uploaded and displayed in a web or mobile app every two minutes. This dual monitoring approach eliminates manual counting errors and enhances accuracy, efficiency, and automation. Users will be able to receive timely updates and notifications via the app, improving productivity and reducing labor costs.

7.4 Limitations:

- Battery or Power Constraints: For portable versions, maintaining both LCD display operation and continuous Wi-Fi-based cloud updates may drain the battery quickly, especially in high-volume operations.
- Data Security and Privacy**:** While Firebase allows convenient remote access, transmitting and storing sensitive production data in the cloud may expose it to unauthorized access if not properly secured. Strong encryption and authentication measures are required for the mobile app and database.
- Sensor Calibration**:** Accurate counting depends on properly calibrated ultrasonic sensors. Misalignment or poor tuning could result in false counts, which will be reflected both on the LCD and the cloud data.
- Cloud Dependency**:** The system's ability to update the mobile app every 2 minutes relies on stable internet connectivity. Network interruptions may cause delays in count updates or missed logs.

Chapter-8

Conclusions and Future Work

8.1 Conclusions:

This project successfully developed an IoT-driven automated paper counting system using ultrasonic sensors, the ESP32 microcontroller, and cloud integration with Firebase. The primary goal of automating the paper counting process was achieved by detecting sheets as they pass through a roller, updating the count in real time, and uploading the data to the cloud for easy remote access. The system significantly improves efficiency, reduces manual errors, and eliminates the need for continuous supervision in high-volume operations. A 16x2 LCD display provides real-time local feedback of the current sheet count, allowing immediate monitoring by on-site operators. Additionally, a mobile/web app interface was developed, which synchronizes with the cloud to display the latest uploaded count every two minutes, offering a user-friendly platform for remote tracking and decision-making.

The key innovation in this project lies in the seamless integration of IoT technology with cloud-based storage and cross-platform interfaces, providing real-time updates locally and periodic cloud-based synchronization for remote users. This dual-feedback mechanism sets it apart from traditional manual counting methods by enhancing visibility and responsiveness. Furthermore, the system is cost-effective due to the use of affordable, off-the-shelf components while still maintaining a high level of accuracy and reliability. Its scalable and adaptable design makes it suitable for various operational environments, from small businesses to industrial setups. Overall, the project meets its objectives, offering a practical and efficient solution for automating paper counting and improving workflow management.

8.2 Future Work:

While the current project has met its primary goals, there are several areas for potential improvement. Future developments could focus on enhancing sensor accuracy, especially in environments with varying paper sizes or materials, to further ensure reliable counting. Incorporating the ESP32-CAM and transitioning the system into a machine learning (ML)-based project could enable identification and classification of different types of sheets—such as coloured, glossy, or textured—passing through the roller, allowing differentiated counting and improved reporting. For improved portability, a battery-powered version could be developed, extending usability to compact or mobile setups. Further, integrating advanced data analytics into the cloud platform would enable trend analysis and performance metrics, supporting informed business decisions. Finally, linking the system to existing industrial machinery would facilitate seamless data exchange, making the solution more scalable, intelligent, and adaptable to a wider range of operational environments.

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Appendix

1. Data sheets:

- **Ultrasonic sensor:**

HC-SR04

2.3、Module pin definitions

Types	Pin Symbol	Pin Function Description
HC-SR04	VCC	5V power supply
	Trig	Trigger pin
	Echo	Receive pin
	GND	Power ground

2.4、Electrical parameters

Electrical Parameters	HC-SR04 Ultrasonic Module
Operating Voltage	DC-5V
Operating Current	15mA
Operating Frequency	40KHZ
Farthest Range	4m
Nearest Range	2cm
Measuring Angle	15 Degree
Input Trigger Signal	10us TTL pulse
Output Echo Signal	Output TTL level signal, proportional with range
Dimensions	45*20*15mm

- **LCD display 16x2 with I2C module:**

Datasheet
I2C 1602 Serial LCD Module

Product features:

The I2C 1602 LCD module is a 2 line by 16 character display interfaced to an I2C daughter board. The I2C interface only requires 2 data connections, +5 VDC and GND to operate

For in depth information on I2C interface and history, visit: http://www.wikipedia/wiki/I2c

Specifications:

	2 lines by 16 character
I2C Address Range	0x20 to 0x27 (Default=0x27, addressable)
Operating Voltage	5 Vdc
Backlight	White
Contrast	Adjustable by potentiometer on I2c interface
Size	80mm x 36mm x 20 mm
Viewable area	66mm x 16mm

Power:

The device is powered by a single 5Vdc connection.

Pinout Diagram:

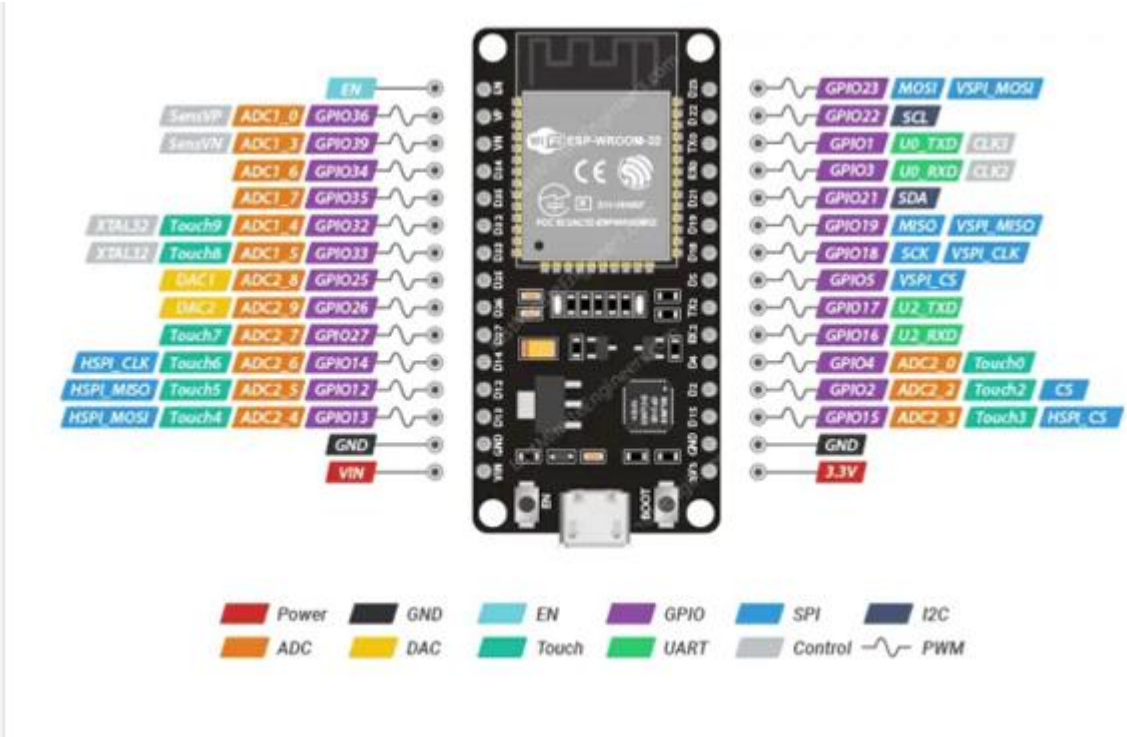
Pin/Control Descriptions:

Pin #	Name	Type	Description
1	GND	Power	Supply & Logic ground
2	VCC	Power	Digital I/O 0 or RX (serial receive)
3	SDA	I/O	Serial Data line
4	SCL	CLK	Serial Clock line
A0	A0	Jumper	Optional address selection A0 - see below
A1	A1	Jumper	Optional address selection A1 - see below
A2	A2	Jumper	Optional address selection A2 - see below
Backlight		Jumper	Jumpered - enable backlight, Open - disable backlight
Contrast		Pot	Adjust for best viewing

Addressing:

A0	A1	A2	Address
Open	Open	Open	0x27
Jumper	Open	Open	0x26
Open	Jumper	Open	0x25
Jumper	Jumper	Open	0x24
Open	Open	Jumper	0x23
Jumper	Open	Jumper	0x22
Open	Jumper	Jumper	0x21
Jumper	Jumper	Jumper	0x20

2. IC Pin Configurations:



3. Developed code:

```
#include <FirebaseESP32.h>
#include <WiFi.h>
#include <Wire.h>
#include <LiquidCrystal_I2C.h>

#define WIFI_SSID "Pixel 9 Pro"
#define WIFI_PASSWORD "hihihihi"

FirebaseConfig config;
FirebaseAuth auth;
FirebaseData firebaseData;
```

```
#define FIREBASE_HOST "https://sheet-counting-machine-default-rtdb.asia-southeast1.firebasedatabase.app/"
#define FIREBASE_AUTH "AIzaSyAJ7kLEZR6pc5A4yPcQ-QNr2u8DosaiSro"

#define TRIG_PIN 5
#define ECHO_PIN 18
#define LED_PIN 2

const int DIST_THRESHOLD_MIN = 7;
const int DIST_THRESHOLD_MAX = 12;
const int AVERAGE_READS = 3;

const unsigned long CHECK_INTERVAL = 20;
const unsigned long MAX_SHEET_DURATION = 2000;
const unsigned long COOLDOWN_PERIOD = 3000;
const unsigned long JAM_CLEAR_DELAY = 10000;
const unsigned long LED_BLINK_DURATION = 5000;
const unsigned long LED_BLINK_INTERVAL = 250;
const unsigned long MAX_DURATION = 120000;

int sheetCount = 0;
bool sheetInRange = false;
bool jammed = false;
bool jammedSheetCounted = false;

bool ledBlinking = false;
bool ledState = false;

unsigned long jamClearedTime = 0;
```

```
unsigned long lastCheckTime = 0;
unsigned long sheetDetectedTime = 0;
unsigned long lastCountTime = 0;

unsigned long ledBlinkStart = 0;
unsigned long lastBlinkToggle = 0;
unsigned long startTime;

float backgroundDistance = 0;
const float DETECTION_THRESHOLD_CM = 2.0;

LiquidCrystal_I2C lcd(0x27, 16, 2);

void setup() {
  Serial.begin(115200);

  pinMode(TRIG_PIN, OUTPUT);
  pinMode(ECHO_PIN, INPUT);
  pinMode(LED_PIN, OUTPUT);
  digitalWrite(LED_PIN, LOW);

  WiFi.begin(WIFI_SSID, WIFI_PASSWORD);
  while (WiFi.status() != WL_CONNECTED) {
    delay(500);
  }

  config.host = FIREBASE_HOST;
  config.signer.tokens.legacy_token = FIREBASE_AUTH;
  Firebase.begin(&config, &auth);
```

```
  Firebase.reconnectWiFi(true);

  startTime = millis();
  backgroundDistance = getAverageDistance(AVERAGE_READS);

  lcd.init();
  lcd.backlight();
  lcd.clear();
  lcd.setCursor(0, 0);
  lcd.print("Sheets:");
  lcd.setCursor(0, 1);
  lcd.print(sheetCount);
}

void loop() {
  unsigned long currentMillis = millis();

  if (currentMillis - startTime > MAX_DURATION) {
    Firebase.setInt(firebaseData, "/final_a4_sheet_count", sheetCount);
    while (true);
  }

  if (ledBlinking) {
    if (currentMillis - ledBlinkStart >= LED_BLINK_DURATION) {
      ledBlinking = false;
      digitalWrite(LED_PIN, LOW);
    } else if (currentMillis - lastBlinkToggle >= LED_BLINK_INTERVAL) {
      ledState = !ledState;
      digitalWrite(LED_PIN, ledState);
```

```
      lastBlinkToggle = currentMillis;
    }
  }

  if (currentMillis - lastCheckTime >= CHECK_INTERVAL) {
    lastCheckTime = currentMillis;
    float avgDistance = getAverageDistance(AVERAGE_READS);
    bool sheetDetectedNow = (backgroundDistance - avgDistance >
DETECTION_THRESHOLD_CM);

    if (jammed && !sheetDetectedNow) {
      if (jamClearedTime == 0) jamClearedTime = currentMillis;
      if (currentMillis - jamClearedTime >= JAM_CLEAR_DELAY) {
        jammed = false;
        jammedSheetCounted = false;
        jamClearedTime = 0;
      }
    } else if (jammed && sheetDetectedNow) {
      jamClearedTime = 0;
      return;
    }

    if ((currentMillis - lastCountTime < COOLDOWN_PERIOD) || jammed) {
      return;
    }

    if (sheetDetectedNow) {
      if (!sheetInRange) {
        sheetInRange = true;
```

```
      sheetDetectedTime = currentMillis;
      lastCountTime = currentMillis;
      sheetCount++;
      Firebase.setInt(firebaseData, "/a4_sheet_count", sheetCount);
      lcd.clear();
      lcd.setCursor(0, 0);
      lcd.print("Sheets:");
      lcd.setCursor(0, 1);
      lcd.print(sheetCount);
    }
  } else {
    if (sheetInRange) sheetInRange = false;
  }

  if (sheetInRange && (currentMillis - sheetDetectedTime >
MAX_SHEET_DURATION)) {
    if (!jammed) {
      jammed = true;

      if (!jammedSheetCounted) {
        sheetCount++;
        jammedSheetCounted = true;
        lastCountTime = currentMillis;
        Firebase.setInt(firebaseData, "/a4_sheet_count", sheetCount);
        lcd.clear();
        lcd.setCursor(0, 0);
        lcd.print("Sheets:");
        lcd.setCursor(0, 1);
        lcd.print(sheetCount);
```

```
        ledBlinking = true;
        ledBlinkStart = currentMillis;
        lastBlinkToggle = currentMillis;
        ledState = true;
        digitalWrite(LED_PIN, HIGH);
      }
    }
  }
 }
}
float getAverageDistance(int samples) {
 long total = 0;
 for (int i = 0; i < samples; i++) {
  total += measureDistance();
  delay(5);
 }
 return total / float(samples);
}

TRIG long measureDistance() {
 digitalWrite(_PIN, LOW);
 delayMicroseconds(2);
 digitalWrite(TRIG_PIN, HIGH);
 delayMicroseconds(10);
 digitalWrite(TRIG_PIN, LOW);
 long duration = pulseIn(ECHO_PIN, HIGH, 30000);
 return duration * 0.034 / 2;
}
```